Every Site Hears the Same

The Loop Equality Is a Theorem

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Paper #102 · Version 3 · 11 November 2026 (Versions 1 and 2: 7 and 10 November 2026) · DICTO DOI: 10.5281/zenodo.24037664 · Licence CC BY-SA 4.0 · Data, scripts and checks: https://github.com/DICTOR-Master/kaleidohedra/tree/09c6a9d4/studies/2026-11-07-loop-equality (fixed commit) · Part of DICTO: Collected Geometric Works (DICTO DOI: 10.5281/zenodo.23345699)
The Loop Equality surfaced in DICTO's study of stackings (#74): the loop counts computed there came out identical on every stacking to length 34, which neither DICTO nor Claude had expected; #74 stated it as a conjecture for all lengths. Proof, checks and drafting with Claude (Anthropic).

Version 3 corrects the account of the origin (the equality surfaced, unexpected, in the loop counts computed for DICTO's study #74; it was not foreseen) and the link to the data, which now points to a fixed commit. No result changes.

Version 2 corrects the notes on methods: the names of our computers and private file paths are removed, with small wording fixes. No result changes.

Stand at one atom of a crystal and walk from atom to atom along the bonds, never turning straight back, until you are home again. How many such round trips are there of each length? In diamond and in its hexagonal cousin lonsdaleite the atoms are packed differently, and further neighbours are not even equally many. Yet in DICTO's study of stackings (#74) the round trips came out the same, to the last digit, for every stacking tried, to everyone's surprise. Here we prove it for every stacking, every atom and every length.

Abstract. The tetrahedral networks of the close-packed stackings (diamond 3C, lonsdaleite 2H, the polytypes 4H, 6H, 9R, and every other stacking, periodic or not) differ in their shells of neighbours from the third shell on. In DICTO's study of stackings (#74) their closed non-backtracking loops turned out not to differ: the numbers of loops of each length from 6 to 34 per site were identical on five stackings, and #74 stated the **Loop Equality** as a conjecture for all lengths. **Theorem 1.** On the tetrahedral network of every close-packed stacking, at every site, the number of closed non-backtracking loops of length m is the number for diamond, for every m: 24, 48, 480, 3432, 25704, … for m = 6, 8, 10, 12, 14, …, given by the Ihara–Bass formula from the Domb numbers. The proof is short. In Fourier space over the layers, the stacking only decides whether a step across a bond layer uses a triangle of three bond directions or its point mirror, and a closed walk crosses every bond layer as often upwards as downwards, so the two choices cancel; non-backtracking loops are then reduced to ordinary walks between the two ends of a bond. The closed-walk part was known (Nagle 1966 for ice; Thorpe 1972 for all periodic polytypes). New here are the loops, the statement at every single site, aperiodic stackings, and two corollaries: the same holds on the pyrochlore network of every stacking (spin ice), and on the sphere packing of every close-packed stacking (fcc, hcp, …). A check of exact integer counts on 42 stackings, including 30 random ones, Fibonacci approximants and the aperiodic Fibonacci stacking, agrees at all 686 sites tested.

1. The networks and the theorem

The network of a stacking. Let Λ be the triangular lattice of unit spacing, spanned by a_1 and a_2, and $t = (a_1 + a_2)/3$; the three close-packed positions A, B, C are Λ, $\Lambda + t$, $\Lambda + 2t$. A stacking is a sequence $\sigma = (\sigma_k)$, $k \in \mathbb{Z}$, of letters with $\sigma_{k+1} \neq \sigma_k$, periodic or not. Its tetrahedral network G_σ, as #74 builds it (loops2.py), has at each bilayer k a lower and an upper atom (k, x, 0) and (k, x, 1) at every point x of position σ_k. Each lower atom is bonded to the upper atom above it, and each upper atom (k, x, 1) to the three lower atoms (k+1, y, 0) of the next bilayer with $|y - x| = 1/\sqrt{3}$. Every atom has four bonds. With ABC repeated, G_σ is the diamond network (3C); with AB, lonsdaleite (2H); with ABAC, ABCACB and ABABCBCAC, the polytypes 4H, 6H and 9R. The pyrochlore network of the stacking (DICTO Janus c/h Pyrochlore, #60) has a site at the midpoint of every bond, two sites being joined when their bonds share an atom; it is the line graph of G_σ.

Loops. A closed non-backtracking loop of length m at a site v is a walk $v = v_0, v_1, \ldots, v_m = v$ along bonds that never turns straight back ($v_{i+1} \neq v_{i-1}$), including at the start ($v_1 \neq v_{m-1}$). These are the loops counted in #74 (the trace of the Hashimoto edge matrix, per site). We write $t_m(v)$ for their number and $W_m(v, w)$ for the number of all walks of length m from v to w.

> **Theorem 1 (the Loop Equality, the conjecture of #74).** For every stacking σ, every site v of G_σ and every $m \geq 1$, $t_m(v)$ equals the number for diamond. Explicitly,

$$\Sigma_{m\geq 1}\, t_m\, u^m/m = -\log(1 - u^2) + \Sigma_{j\geq 1}\, (1/j)\, \Sigma_{i=0..j}\, C(j, i)\, W_i\, u^i\, (-3u^2)^{j-i},$$

where W_i is the number of closed walks of length i at a site of diamond: 0 for odd i and the Domb number (OEIS A002895: 1, 4, 28, 256, 2716, ...) for $i = 2n$. The loop numbers are 0 for odd m and for m = 2, 4, and, for m = 6, 8, ..., 34, the values 24, 48, 480, 3432, 25704, 194256, 1509072, 11820000, 93722376, 750276792, 6056400480, 49241424120, 402912569544, 3315331754256, 27416593190640 that DICTO computed in #74.

The same holds for all closed walks ($W_m(v, v)$ equals W_m at every site of every stacking) and for walks between the two ends of any bond or any path of two bonds (Lemma 2). Two corollaries follow with the same proof.

> **Corollary 2 (spin ice).** On the pyrochlore network of every stacking, at every site and for every length, the numbers of closed walks and of closed non-backtracking loops equal those of the pyrochlore network of diamond (lengths 1 to 14: walks 0, 6, 12, 78, 300, 1512, ...; loops 0, 0, 12, 12, 0, 276, 756, 2988, ...).

> **Corollary 3 (sphere packings).** In the packing of equal spheres of every close-packed stacking (fcc, hcp, dhcp, ...), joining touching spheres, at every sphere and for every length, the numbers of closed walks and of closed non-backtracking loops equal those of fcc (walks: OEIS A002899; loops of length 3 to 12: 48, 264, 1680, 17016, 151872, 1401720, 13238400, 126357120, 1221557040, 11927724720).

The networks themselves are not the same: the numbers of atoms 3 to 8 bonds away differ between sites by up to 5% (Figure 2a). Theorem 1 says that the loops, which these differences would seem to affect, do not see them.

2. Proof

Levels. Give the atom (k, x, u) the level $\ell = 2k + u$. Every bond joins neighbouring levels. Between levels 2k and 2k+1 the bonds are the vertical bonds inside bilayer k, one per atom; between 2k+1 and 2k+2 lies the *bond layer* k, three bonds per atom. Write $D = \{t, t - a_1, t - a_2\}$, the three shortest vectors of $\Lambda + t$. A bond of bond layer k goes from (k, x, 1) to (k+1, x + d, 0) with $d \in \varepsilon_k D$, where $\varepsilon_k = +1$ if $\sigma_{k+1} - \sigma_k \equiv 1$ and $\varepsilon_k = -1$ if $\sigma_{k+1} - \sigma_k \equiv 2 \pmod 3$, since the shortest vectors of $\Lambda - t$ are $-D$ (Figure 1a). The stacking enters only through the signs ε_k. Let $g(\theta) = \Sigma_{d \in D}\, e^{i\theta \cdot d}$ for $\theta \in \mathbb{R}^2$, and let $\int$ denote the normalised integral over the torus $\mathbb{R}^2/\Lambda^*$, where $\Lambda^* = \{\theta : \theta \cdot \lambda \in 2\pi\mathbb{Z} \text{ for all } \lambda \in \Lambda\}$, so that $\int e^{i\theta \cdot \lambda} = 1$ for $\lambda = 0$ and 0 for the other $\lambda \in \Lambda$.

> **Lemma 1.** For sites v = (k, x, u) and w = (k′, y, u′), $W_m(v, w) = \Sigma_P \int e^{-i\theta \cdot (y-x)} \Pi_s \varphi_s(\theta)$, the sum over the walks P of length m on $\mathbb{Z}$ from level 2k+u to level 2k′+u′, the product over their steps s, with $\varphi_s = 1$ for a step inside a bilayer, $g(\varepsilon_j\theta)$ for an upward step across bond layer j and $g(-\varepsilon_j\theta)$ for a downward one.

Proof. A walk in G_σ is its sequence of levels P together with the in-plane displacement d_s of each step, chosen from $\{0\}$, $\varepsilon_j D$ or $-\varepsilon_j D$ according to the step. The choices are independent: from every atom at the start of a step the available displacements are exactly these, since every layer is a translate of Λ and the bond rule is the same at every point. The walk ends at w exactly when $\Sigma\, d_s = y - x$. For every choice, $\Sigma\, d_s - (y - x)$ lies in Λ (both ends of the walk lie in the same position class as w), so the number of choices with $\Sigma\, d_s = y - x$ is $\int e^{-i\theta \cdot (y-x)} \Pi_s (\Sigma_d\, e^{i\theta \cdot d})$. ∎

Two facts about g. (i) $g(-\theta)$ is the complex conjugate $\bar{g}(\theta)$, so $g(\varepsilon\theta)g(-\varepsilon\theta) = |g(\theta)|^2$ for $\varepsilon = \pm 1$. (ii) $|g(\theta)|^2 = 3 + \Sigma_\lambda\, e^{i\theta \cdot \lambda}$, the sum over the six shortest vectors λ of Λ (the differences of two elements of D); so $|g|^2$ is invariant under the twelve symmetries of Λ fixing 0, which permute the six shortest vectors transitively and preserve the torus integral. Hence, for any function F of $|g|^2$, the integral $J_F = \int F\, e^{i\theta \cdot \lambda}$ is the same for all six shortest λ.

> **Lemma 2.** The numbers $W_m(a, b)$ do not depend on the stacking σ, and are equal to the corresponding numbers of diamond, when (a) a = b; (b) a and b are the two ends of a bond; (c) a and b are the two ends of a path of two bonds.

Proof. (a) A closed level path P crosses each bond layer j as often upwards as downwards, n_j times each. By Lemma 1 and fact (i) its contribution is $\int |g|^{2n(P)}$ with $n(P) = \Sigma\, n_j$, whatever the signs ε_j. The level paths and their numbers n(P) depend only on m and on the parity of the start level; the reflection $\ell \mapsto 4k + 1 - \ell$ exchanges levels 2k and 2k+1 and maps bilayers to bilayers and bond layers to bond layers, so the parity does not matter either (Figure 1b). (b) For a vertical bond, the level paths from 2k to 2k+1 still cross every bond layer equally often both ways, and $y - x = 0$: the count is $\Sigma_P \int |g|^{2n(P)}$. For a bond of bond layer k from (k, x, 1) to (k+1, x + d, 0), $d = \varepsilon_k\delta$ with $\delta \in D$, the level paths cross bond layer k once more upwards than downwards and all others equally often, so the integrand is $|g|^{2n(P)}\, g(\varepsilon_k\theta)\, e^{-i\theta \cdot d} =$

$|g|^{2n(P)}$ $(1 + e^{i\theta\cdot\lambda} + e^{i\theta\cdot\lambda'})$, with λ, $\lambda' = \varepsilon_k(\delta' - \delta)$ two shortest vectors of Λ. By fact (ii) the integral is $\int |g|^{2n(P)} + 2J$, independent of ε_k and δ. (c) Let b be the middle atom, say an upper atom (k, x, 1) (a lower one is the mirror case under the reflection of (a)). If a and c both lie above b, they lie on one level, and c − a is a shortest vector of Λ: the integrand is $|g|^{2n(P)}$ $e^{-i\theta\cdot(c-a)}$, of integral J. If a lies below b and c above, the level paths cross bilayer k and bond layer k once more upwards, and the integrand is that of a bond in (b). In every case the count is given by a formula in which σ does not appear; diamond is one of the stackings, so the count is diamond's. ∎

Diamond's symmetries carry every bond to every other and every path of two bonds to every other (the 24 symmetries fixing an atom permute its four bonds in all 24 ways). So in Lemma 2 the counts of (b) are one number for all bonds of all stackings, and those of (c) one number for all paths of two bonds.

> **Proposition (the closed walks of diamond as a chain).** $W_{2n} = \Sigma_{a=0..n}\, C(n, a)^2\, H(a)$, where $H(a) = \Sigma_{i+j+k=a}\, (a!/(i!\, j!\, k!))^2$ is the number of closed walks of length 2a on the honeycomb net (OEIS A002893).

Proof. By (a), $W_{2n} = \Sigma_P\, H(n(P))$, since $\int |g|^{2a} = H(a)$: $g^a = \Sigma\, (a!/(i!j!k!))\, e^{i\theta\cdot(id_1+jd_2+kd_3)}$ and the exponents determine (i, j, k). The number of closed level paths of length 2n with n(P) = a is the constant term of the 2×2 transfer matrix $M(\varphi)^{2n}$, $M = [[0,\ 1 + \sqrt{z}\, e^{-i\varphi}],\ [1 + \sqrt{z}\, e^{i\varphi},\ 0]]$ (weight $\sqrt{z}$ per crossing of a bond layer), and $M^2 = |1 + \sqrt{z}\, e^{i\varphi}|^2\, I$, whose n-th power has z^a-coefficient $C(n, a)^2$ at φ-frequency 0. ∎ This is Peter Bala's comment on A002895 (the Domb numbers are the row sums of the cube of the triangle $C(n, k)^2$), read as walks: line, then layers.

From walks to loops. Let H be any graph in which every vertex has q + 1 neighbours, finite or not. Let $A_j(x, y)$ be the number of non-backtracking walks of length j from x to y (with no condition at the ends). They satisfy $A_0 = I$, $A_1 = A$, $A_2 = A^2 - (q+1)I$ and $A_{j+1} = A\, A_j - q\, A_{j-1}$ for $j \geq 2$, where A is the adjacency matrix: so $A_j(x, y)$ is a fixed integer combination of the walk numbers $W_i(x, y)$, $i \leq j$.

> **Lemma 3.** For a bond e directed from a to b, let $\tau_m(e)$ be the number of closed non-backtracking loops of length m whose first step is e. Then $\tau_m(e)$ is an integer combination, with coefficients depending only on q and m, of $W_i(a, a)$, $W_i(b, b)$ and $W_i(a, b)$, $0 \leq i \leq m$.

Proof. Let $\bar{e}$ be e reversed, $P_k(e, y)$ the number of non-backtracking walks of length k with first step e that end at y, and α_k, β_k the numbers of those whose last step is e, respectively $\bar{e}$. A walk with first step e is e followed by a non-backtracking walk from b whose first step is not $\bar{e}$, so $P_1(e, y) = [y = b]$ and $P_k(e, y) = A_{k-1}(b, y) - P_{k-1}(\bar{e}, y)$: unwinding, $P_k(e, y) = \Sigma_{i=0..k-1}$ $(-1)^i\, A_{k-1-i}(x_i, y)$ with $x_i = b$ for even i and a for odd i. Removing the last step, $\alpha_1 = 1$, $\beta_1 = 0$, $\alpha_k = P_{k-1}(e, a) - \beta_{k-1}$ and $\beta_k = P_{k-1}(e, b) - \alpha_{k-1}$. Finally a loop with first step e is a walk with first step e back to a whose last step is not $\bar{e}$: $\tau_m(e) = P_m(e, a) - \beta_m$. Only $A_j(x, y)$ with x, $y \in \{a, b\}$ occur. ∎

Proof of Theorem 1. G_σ has four bonds at every atom (q = 3). By Lemma 3, $\tau_m(e)$ for a bond e of G_σ is a fixed combination of walk numbers that Lemma 2 (a), (b) shows to be diamond's; so $\tau_m(e)$ is diamond's value, which is the same for all directed bonds of diamond. Then $t_m(v) = \Sigma\, \tau_m(e)$ over the four bonds e leaving v is the diamond value at every site. For the formula, take the quotient X of diamond by a large multiple of its lattice of translations, so that no walk of length $\le m$ wraps round. X is finite with every vertex of degree 4 and $|E| - |V| = |V|$. The Ihara–Bass formula, $\Sigma_m N_m u^m/m = -(|E| - |V|) \log(1 - u^2) - \log\det(I - uA + 3u^2 I)$, with $N_m = |V|\, t_m$ and $\log\det = -\Sigma_j \operatorname{tr}(uA - 3u^2)^j/j$, $\operatorname{tr} A^i = |V|\, W_i$, gives the stated series after division by |V|. ∎

Proof of Corollary 2. Let S be the incidence matrix of atoms and bonds. The pyrochlore network has adjacency matrix $S^TS - 2I$, and $(S^TS)^j = S^T(SS^T)^{j-1}S$ with $SS^T = A + 4I$. So the number of walks of length m between two pyrochlore sites e and f is a fixed combination of $W_i(x, y)$ with x an end of bond e and y an end of bond f. For e = f these are pairs of type (a) and (b) of Lemma 2; for neighbouring sites e = {a, b} and f = {b, c} they are of types (a), (b) and (c). Lemma 3 with q = 5 then gives the loops, as in the proof of Theorem 1; diamond's pyrochlore network carries every directed pair of neighbouring sites to every other, by the symmetry noted after Lemma 2. ∎

Proof of Corollary 3. The same, with one level per close-packed layer. A step inside a layer contributes $c(\theta) = \Sigma_\lambda\, e^{i\theta\cdot\lambda}$ over the six shortest λ, a real function invariant under the symmetries of Λ; a step to the layer above contributes $g(\varepsilon_k\theta)$ and to the layer below $g(-\varepsilon_k\theta)$. Closed walks give products of c and $|g|^2$, and so does a step between touching spheres, up to the factor treated in Lemma 2 (b). fcc carries every pair of touching spheres to every other, so Lemma 3 with q = 11 gives the loops. ∎

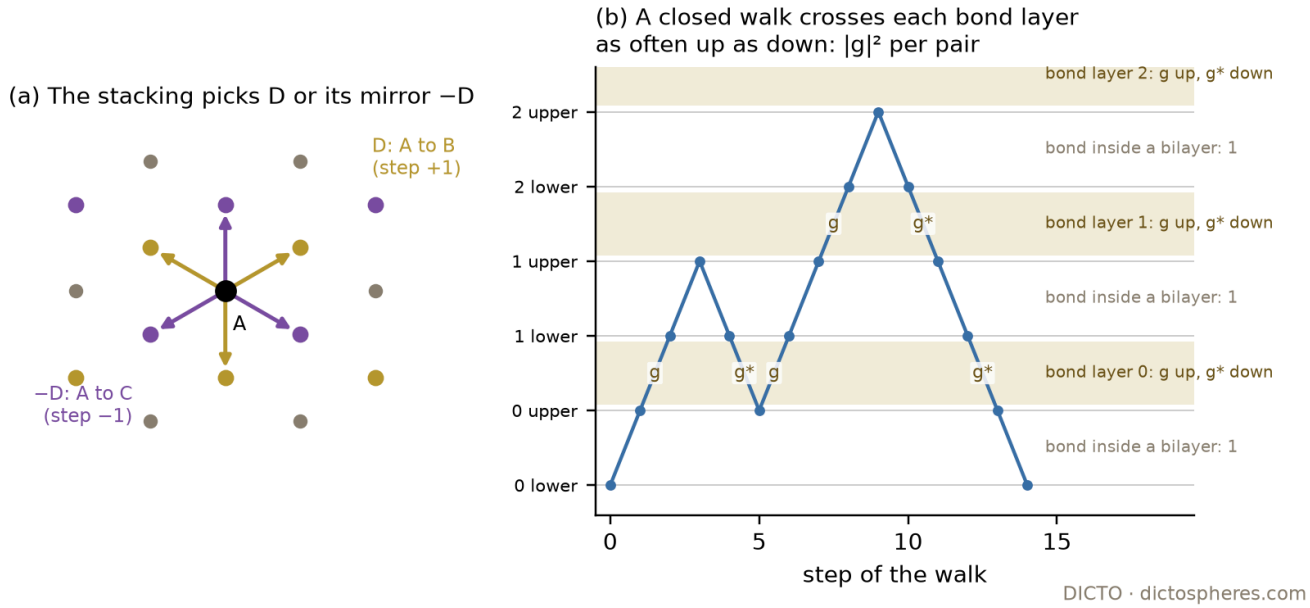


Figure 1. The proof in two pictures. (a) From an atom at an A position, the three bonds to the next bilayer go to B positions (gold, the triangle D) or to C positions (violet, its point mirror −D); the stacking decides which. (b) The levels of a closed walk: inside a bilayer a step counts 1; across a bond layer (gold bands) an upward step counts g and a downward step its conjugate g*, whatever the triangle. A closed walk crosses each band as often up as down, so it counts $|g|^2$ per pair, the same on every stacking.

3. Checks

The proof was checked against exact integer counts. The networks were built from their bond rule (that of loops2.py and oeis_checks.py) on windows of N × N in-plane cells with N larger than the walk length, so that no walk can wrap round, and with enough bilayers above and below the tested sites that no walk reaches the window's edge. The tested sites are all sites of one period, or of 12 consecutive bilayers when the period is longer or the stacking aperiodic.

Stackings	What was counted, at every tested site	Result
42: 3C, 2H, 4H, 6H, 9R; 30 random periodic stackings, periods 4 to 12 (hexagonality 0.17 to 0.80); Fibonacci approximants of periods 3, 5, 8, 13, 21; the aperiodic Fibonacci stacking; one random aperiodic stacking. 686 sites.	closed walks, lengths 0 to 30; walks between the ends of every bond and of every path of two bonds, lengths 0 to 30; closed non-backtracking loops, lengths 1 to 24	all equal to diamond (Domb numbers; Ihara–Bass values)
19 of these (2H to rand07, the approximants, both aperiodic ones)	pyrochlore network: closed walks and loops, lengths 1 to 14, every site	all equal to 3C
the same 19, as sphere packings	closed walks and loops, lengths 1 to 12, every sphere	all equal to fcc
diamond	the chain formula of the Proposition, n = 0 to 15	equal to the Domb numbers
controls	atoms 3 to 6 bonds away: 3C 24, 42, 64, 92 and 2H 25, 44, 67, 96 (OEIS A008253, A008264); per site, n = 3 to 8, up to 4.9% from diamond	the networks differ, as they should

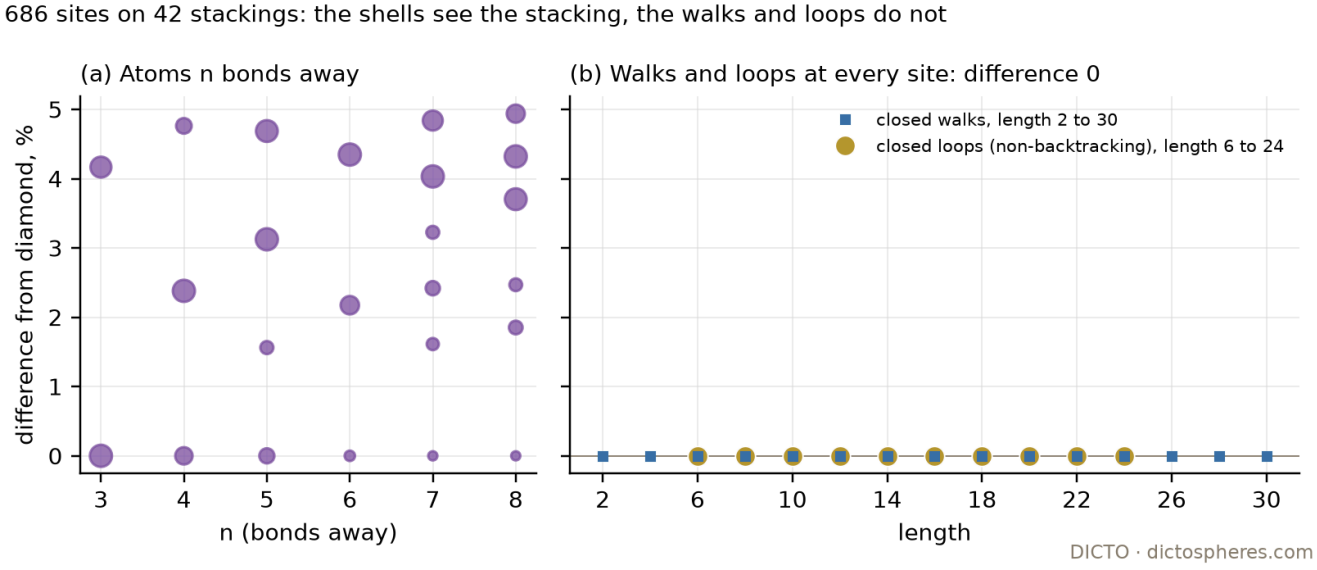


Figure 2. All 686 tested sites of the 42 stackings. (a) The number of atoms n bonds away, against diamond: the sites differ by up to 4.9% (dot size = number of sites). (b) Closed walks of length 2 to 30 (blue) and closed non-backtracking loops of length 6 to 24 (gold) at the same sites: the difference from diamond is exactly 0 at every site and every length.

4. Where the result stops

(1) Theorem 1 is about counts, not shapes: boat-shaped six-rings occur only where the stacking is hexagonal, 3h/2 per atom against 2 − 3h/2 chairs (#74, the count after its Theorem 1), yet the numbers of loops agree. (2) Lemma 2 covers pairs of sites at most two bonds apart. For sites further apart the networks differ, which is why the shells differ; the equality is spectral (every site has the same local spectrum), not an isomorphism. 4H and 6H

are thus infinite networks whose sites are not all alike but have equal numbers of closed walks: walk-regular but not vertex-transitive. (3) The step from loops to the frozen-in entropy of spin ice and of ice, item (2) of #74 Section 5, is unchanged: the entropy depends on how loops overlap and combine, not only on their numbers, and its equality for 3C and 2H is still supported by numerical evidence only (Nagle 1966; Xu, Lin and Zhang 2025). (4) The proof needs every layer to be a full close-packed layer and only nearest-neighbour bonds; it does not cover stackings with faults inside a layer, or networks with longer bonds.

5. What is known, and what is new

Known, credited. Nagle (1966) noticed, for ice, that the generating function of walks returning to the origin is the same for hexagonal and cubic ice (as reported by Isaev, McKay and Zhang 2024, §8.2); Thorpe (1972) proved that the number of closed walks of each length is the same for all polytypes of the fcc–hcp series and of the zincblende–wurtzite series, independent of the starting point, by showing that a simple Hamiltonian has the same eigenvalues (from his abstract). This is Lemma 2 (a) and the closed-walk part of Corollary 3 for periodic stackings. Betteridge (1981) found an exact one-to-one correspondence of the nearest-neighbour s bands of fcc and hcp, and He, Wilson and Ganesh (2024) explained it by a gauge symmetry, one phase per layer, that makes the band structures of all close-packed stackings equal; Lemma 1 with fact (i) is the same mechanism, written for walks. The Ihara–Bass formula and the recursion for non-backtracking walks are standard (Ihara 1966; Hashimoto 1989; Bass 1992; for infinite periodic graphs Clair and Mokhtari-Sharghi 2001, Guido, Isola and Lapidus 2008). The chain formula is Peter Bala's comment on OEIS A002895. **DICTO's earlier work, cited:** the Loop Equality and its counts through length 34 (#74; HexOct book, Chapter 13); the pyrochlore network of every stacking (#60). **New here:** the proof of the Loop Equality for non-backtracking loops (Theorem 1), at every single site and for aperiodic stackings as well as periodic ones; Lemma 2 (b), (c) and Lemma 3, which carry the equality from closed walks to loops site by site; Corollary 2 for the pyrochlore network of every stacking; the loops of Corollary 3. We found no earlier statement that non-backtracking loop counts agree across stackings. #74 said it found no earlier statement that networks of different stacking have equal loop counts; that holds for its non-backtracking loops, but the closed-walk equality is Nagle's and Thorpe's.

Sources checked (7 November 2026). *Reference library*: INDEX.html, cited/MANIFEST.json, papers/MANIFEST.json, expansion/MANIFEST.json and cited/dicto-own.json, searched for Ihara, closed walk, return probability, walk-regular, cospectral, lonsdaleite, polytype, non-backtracking and Domb: hits only in abstracts on polytype energetics (Ising models of close-packed polytypes; Lennard-Jones polytypes; iron LPSO), not on walks; the text of all 259 PDFs, searched for the same terms: hits in Zhu et al. 2026 (lonsdaleite structure references), Görlich 2011 (return probability in causal dynamical triangulations), and the titles of Jenczyk 2024, Ambjørn–Loll 2024, Nguyen–Cava 2020, George et al. 2020, Keller et al. 2023 and the index, none on walk counts. *Web*, read: the OEIS entries A002895 (formulas and comments), A293237 (fcc escape probability; comment that hcp has the same value; reference Ishioka and Koiwa 1978); Crossref records with abstracts of Thorpe 1972 and Betteridge 1981 I and II; He, Wilson and Ganesh, arXiv 2405.15865 v1 (mechanism and references); Isaev, McKay and Zhang, arXiv 2409.04989 v2 (§8.2); Bercu and Montégut, arXiv 2109.08397 (random walks on ice and graphite; no comparison of stackings); OpenAlex list of the 17 works citing Thorpe 1972 (titles; none on non-backtracking walks). *Bibliographic records checked on Crossref, papers not re-read:* Nagle 1966; Ihara 1966; Hashimoto 1989; Bass 1992; Clair and Mokhtari-Sharghi 2001; Guido, Isola and Lapidus 2008. *Records seen, not read:* Ishioka and Koiwa 1978; Lukes and Nix 1973; Koiwa 1977; Wilson, Ganesh and Samokhin, arXiv 2602.18574; Clair 2013 (arXiv 1306.5454); Guido, Isola and Lapidus, arXiv math/0608229; Kempton, arXiv 1603.05553; Kong, Wyman and Xi, arXiv 2401.06022 (walk-regular planar graphs). *Web searches:* closed walks diamond lonsdaleite polytypes; hcp fcc return probability closed walks; Ihara zeta periodic graphs

diamond; non-backtracking walks Ihara lonsdaleite wurtzite; walk-regular cospectral vertices infinite periodic graphs; ice Ih Ic residual entropy loops; Thorpe 1972 polytypes; "Random walks in polytype structures".

Data and code

Study 2026-11-07-loop-equality: loop_checks.py (all checks of Section 3, exact integers; run in two parts, 88 s and 122 s, output data-loop-checks.txt, both parts FINAL ok) and fig102.py (both figures from data-loop-checks.txt). The bond rule is that of studies 2026-10-28-pyrochlore-kagome (loops2.py, #74) and 2026-11-07-oeis-candidates (oeis_checks.py, whose C1 and C2 gave the first hint: closed walks equal to the Domb numbers at every site of 3C, 2H, 4H, 6H to length 30).

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